

Machine Learning and its Applications

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Today's goals



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Module 1

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Machine Learning Fundamentals I: Basic Concepts

Module 2

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Machine Learning Fundamentals II: Model Training and Evaluation

Module 3

▼

Data Preparation

Module 4

▼

Support Vector Machines

Module 5

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Artificial Neural Networks

Module 11 Introduction

✓

Introduction to Module 11's Study

Video • 1 min

Module 11-1 Artificial Neural Network Introduction

✓

Artificial Neural Network Introduction: Introduction

Video • 3 min

✓

Artificial Neural Network Introduction: ANN Structure

Video • 4 min

✓

Materials on ANN fundamentals

Reading • 10 min

Module 11-2 Neural Network Training

✓

Neural Network Training: Forward Propagation

Video • 5 min

✓

Neural Network Training: Backward Propagation

Video • 13 min

✓

Neural Network Training: More on Neural Network Training

Video • 8 min

✓

Materials on Neural Network Training

Reading • 10 min

Module 11-3 Underfitting and Overfitting

✓

Underfitting and Overfitting: Concepts

Video • 7 min

✓

Underfitting and Overfitting: Methods to Improve Overfitting

Video • 8 min

✓

Materials on Underfitting and Overfitting

Reading • 10 min

Module 11-4 ANN Implementation in MATLAB and Case Studies

✓

ANN Implementation in MATLAB and Case Studies: Build an ANN Using MATLAB

Video • 6 min

✓

ANN Implementation in MATLAB and Case Studies: A Case Study on Diabetes Diagnosis

Video • 1 min

✓

Materials on ANN Implementation

Reading • 10 min

Module 11 Tutorials

✓

Tutorial: MATLAB Classification and Regression Learner Apps

Video • 6 min

✓

Tutorial: Classification Learner

Video • 10 min

✓

Tutorial: Export Model

Video • 10 min

Module 11 Recap

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Artificial Neural Networks were the most flexible and expressive models studied in this course, but also the most sensitive to design choices and data quality.

Reflect on the **entire course** and address the following:

1. **Where do neural networks clearly outperform earlier methods** (linear models, kNN, SVM), and why?
2. **Where do neural networks introduce new risks** (overfitting, instability, interpretability, computational cost) that simpler models avoid?
3. In a real engineering or scientific project, how would you **justify choosing a neural network over a simpler model**, and what evidence would you require before trusting its results?
4. How do data preparation, validation strategy, and regularization determine whether a neural network becomes a powerful generalizer or an unreliable black box?

Go to next item

✓ Completed

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